Using Packet Sniffer Simulator in the Class: Experience and Evaluation

Xiaohong Yuan[1], Percy Vega[2], Jinsheng Xu[1], Huiming Yu[1], Yaohang Li[1]

[1]Department of Computer Science
North Carolina A&T State University
Greensboro, NC 27411 U.S.A.
Telephone: (336)3347245
{xhyuan;jsxu;cshmyu;yaohang}@ncat.edu

[2]Everett Consulting Corp.
757 SE 17th St Causeway #249
Fort Lauderdale, FL 33316 U.S.A
Telephone: (954)7325664
percyvega@gmail.com

ABSTRACT

Computer visualization and animation have been used in computer science education to help students better understand the concepts. This paper presents an animated simulator for packet sniffer and its use in a network security course. The animated packet sniffer simulator was designed to demonstrate graphically the security concept packet sniffer and related computer networks concepts. It is geared towards an introductory level computer security course. This tool was used in a network security course, and its effectiveness in teaching and learning was assessed. The evaluation result shows that this animation tool is effective in meeting its learning objectives.

Categories and Subject Descriptors

C.2.0 [Computer-Communication Networks]: General – *security and protection*
I.6.5 [Simulation and Modeling]: Model Development – *modeling methodologies*
K.3.1 [Computers and Education]: Computer Uses in Education - *CAI*

General Terms:

Security, courseware

Keywords

Computer science education, security, packet sniffer, animation

1. INTRODUCTION

Visualization and animation have been used in various fields of computer science education, such as algorithm, computer networks, computer architecture and so on [1-3]. Research has shown that there is a widespread belief that visualization technology positively impacts learning [4].

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This paper presents a software tool that demonstrates the security concept of packet sniffer and related computer networks concepts through animation. It is intended to be used in an undergraduate level computer security course or a computer network course. This tool was used in the “network security” course in Fall 2005 semester, and its impact in teaching and learning was evaluated. A pre-test and post-test was given to the students before and after the tool was introduced to the students. Homework assignment on the tool was given to the students. Students opinions on the tool were also collected.

The following sections will describe the concept of packet sniffer, the packet sniffer simulator we designed, and the evaluation result of the packet sniffer simulator.

2. PACKET SNIFFER

A packet sniffer is normally used by network administrator to monitor and troubleshoot network traffic. It captures all of the data packets going through a given network interface. Packet sniffer can also be used by malicious intruders for illicit purpose such as stealing a user’s password or credit-card number [5]. There exist various commercial and free packet sniffer tools [6, 7, 8]. Normally, a computer in a network would only capture data packets that were intended for that machine and ignore packets that are not intended for it. In order for a computer to capture all packets traversing the network, its network interface needs to be configured into promiscuous mode. A packet sniffer can only capture data packets within a given subnet.

3. THE PACKET SNIFFER SIMULATOR

The packet sniffer simulator was intended to demonstrate how a packet sniffer works progressively. It includes both high level explanation and lower level technical details. Therefore, the packet sniffer simulator was designed to consist of a suite of five demos: direct path, real path, promiscuous mode, packet sniffer and Telnet Over TCP/IP. Demo I to IV progressively demonstrate how a packet sniffer works at a higher level. Demo V intends to depict how a data packet is transmitted at a more in-depth level. It displays a protocol stack and animates the encapsulation and de-encapsulation process.

The packet sniffer simulator was implemented using Macromedia Flash MX Professional Edition. The flash animations can run from a web page as a Flash Applet. They can also run as a standalone application. Macromedia Flash Player, which is freely available, is required for running the packet sniffer simulator. In what follows, the fives demos of the packet sniffer simulator, and the network and security concepts they demonstrate are explained.

3.1 Demo I: Direct Path

Figure 1 shows the user interface that Demo I to IV use. The user interface includes four components:

- The "Demo Sequence" component. It is a textbox at the top-left location of the interface which lists the names of the five demos. The user can select the domo he wants to view by clicking on the demo name.
- The "Description Message" component. It is a text-area at the top-right location of the interface which briefly describes the animation. It is dynamically updated with the animation.
- The "Network Architecture" component. The bottom-right space of the interface displays the network architecture used in the simulation. It includes two subnets connected by a router. One subnet has a star topology, while the other has a bus topology. The computers in each subnet are connected with a hub. The computers are identified with numbers from 0 to 8.
- The "Simulation Data" component. This component locates at the bottom-left space of the interface. Demo I – IV allow the user to interact with the simulator through changing the input data. The input data used in the simulation is the source and destination addresses of a data packet. The user can choose from one of the following options: loading default data from an input file, generating input data randomly or entering data manually. The input data are displayed in the table that has two columns: "From" and "To" columns. They indicate the source and the destination addresses respectively.

Under the two-column data table, a "play" button (indicated by an arrow within a square), and a checkbox "Play continuously" are provided. If the "play" button is clicked with "Play continuously" checkbox unchecked, the demo will show the animation for one data pair. This allows the user to step through the simulation by clicking the "play" button repeatedly. If the "play" button is clicked with "Play continuously" checkbox checked, the animation will run from Demo I through Demo IV sequentially and go through all the data pairs in the data table sequentially in each demo.

Demo I displays the path a data packet goes through to reach the destination. A data packet is represented as an oval, labeled by the source and destination numbers. Figure 1 shows a data packet, Packet 1-8 moving from computer 1 to the hub. It will go through the hub to the left, the router in the middle, the hub to the right, and finally arrive at computer 8. Since hub is used in the two subnets, the real path that Packet 1-8 traverses is not the same as the direct path. Demo II demonstrates the real path of a data packet in the simulated network.

3.2 Demo II: The Real Path

Since a hub is used to connect the computers in each subnet, all traffic can be seen by all the computers in the subnet. The network interface hardware of each computer in the subnet detects the electrical signal and extracts a copy of the frame. It checks the address of each incoming frame to determine whether it should accept the frame. The network interface hardware compares the destination address in the frame to the computer's physical address. If they match, the interface hardware accepts the frame and passes it to the operating system. If they do not match, the interface hardware discards the frame and waits for the next frame to appear [9].

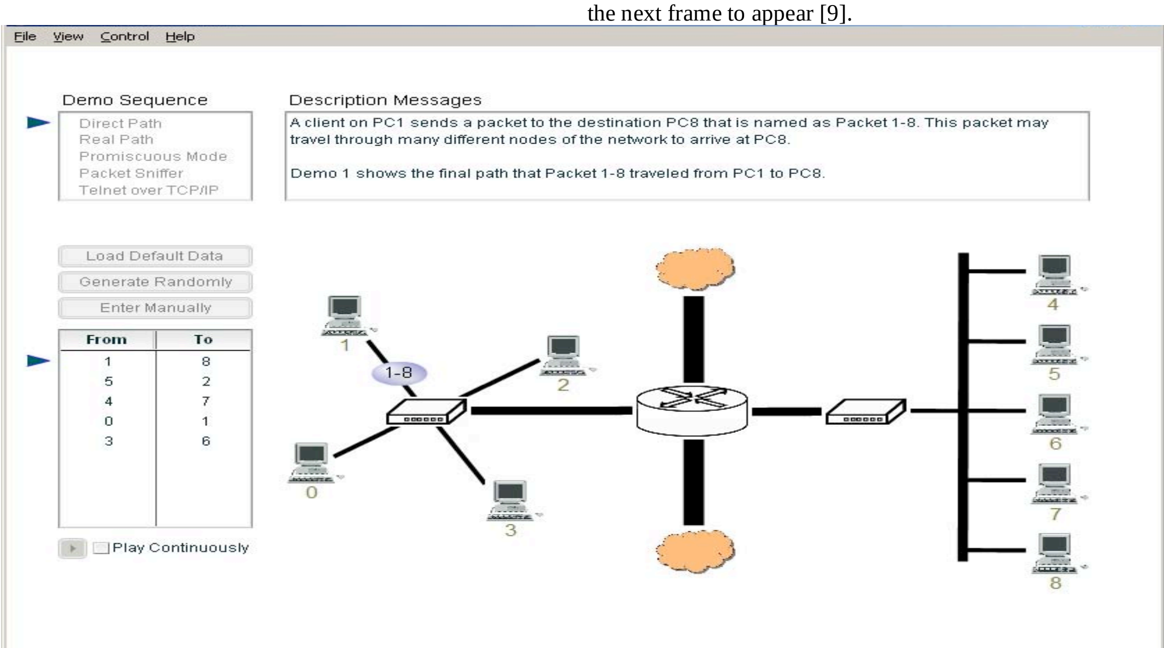

Figure 1. Demo I: Direct Path

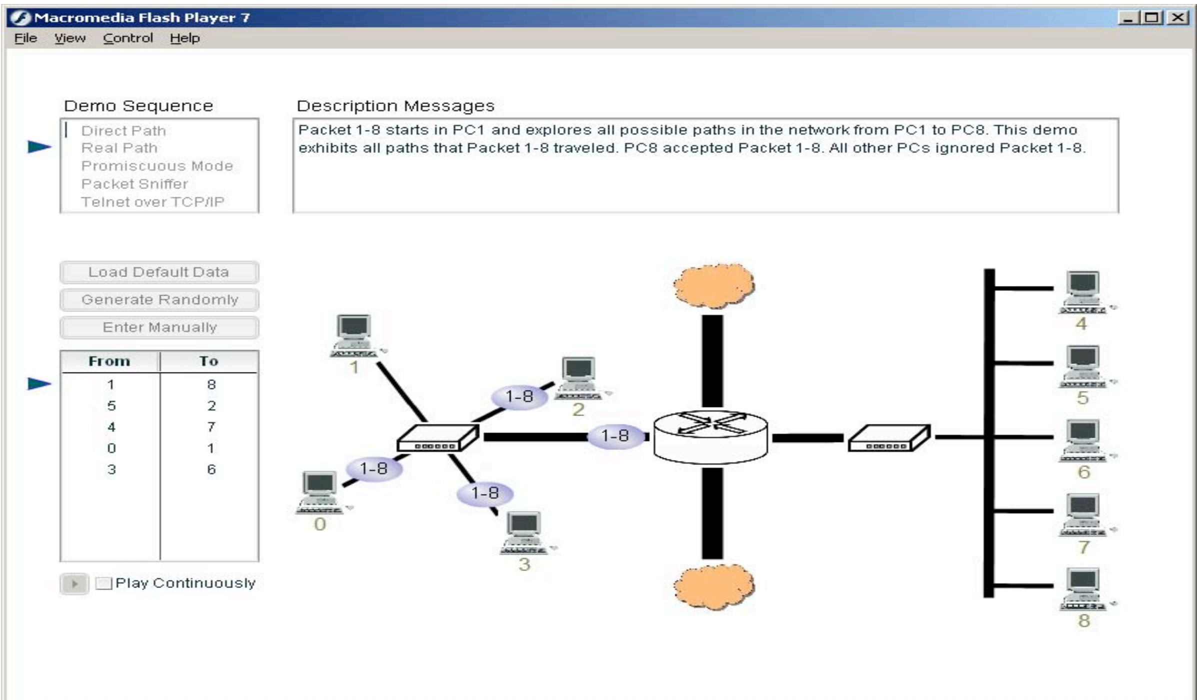

Figure 2. Demo II: The Real Path When A Packet Is Sent by the Source Computer

Figure 2 shows that, after the hub receives Packet 1-8, it broadcasts Packet 1-8 to computer 0, 2 and 3 in the same subnet and the router. The router then forwards Packet 1-8 to the subnet with bus topology. All the computers attached to the bus will receive Packet 1-8, but only computer 8's network interface will accept Packet 1-8. The other computers will discard the data packet. If the hubs in both subnets are replaced by switches or bridges, then the path Packet 1-8 traverses will be the same as in Demo I. By comparing Demo I and Demo II, the concepts of repeater, hub, bridge, switch and router can be reviewed with the students [10].

3.3 Demo III: Promiscuous Mode

Normally, a computer's network interface hardware checks the destination address of each incoming frame to determine whether the frame should be accepted. The network interface discards a frame whose destination address does not match the physical address of the network interface hardware. However, a computer's network interface hardware can be configured by software into promiscuous mode, which overrides the conventional address checking. Once in promiscuous mode, the network interface does not check the destination address of the incoming frame, but accepts all frames. The network interface simply places a copy of each frame in the computer's memory and informs the CPU about the arrival of the frame [9]. This is demonstrated in Demo III by labeling the computer that is configured into promiscuous mode with a "DISCARD" comment indicating that the data packet captured by the computer is simply discarded, since there is no packet sniffer installed on the computer to process the packet. The other computers that are not configured into promiscuous mode are labeled with a "not mine" comment indicating that they have checked the destination address of the frame and have rejected the frame.

3.4 Demo IV: Packet Sniffer

Figure 3 shows the result of an animation when computer 3 is configured into promiscuous mode and also has a packet sniffer installed. Computer 3 captured Package 1-8 and accepted it, even though the packet was not addressed to computer 3. This is indicated by the "mine" comment in red color in Figure 3. In comparison, computer 8 is labeled with "mine" comment in green color since it is the normal destination of Packet 1-8. The packet sniffer then examines the content of the data packet according to its configuration.

3.5 Demo V: Telnet Over TCP/IP

Demo V intends to illustrate how a data packet is transmitted in a network at an in-depth level. It displays a protocol stack and animates the encapsulation and de-encapsulation process. It assumes a Telnet application sending data packets over a network with TCP/IP protocol. Figure 4 represents three computers (PC0, PC1 and PC2) connected to a hub. In each computer a protocol stack of five layers are displayed: application, transport, network, data link and physical layer. The animation shows a data packet generated at the application layer at PC0 being encapsulated while moving down through the protocol stack, and being de-encapsulated while moving up through the protocol stack at PC1 and PC2. During the encapsulation process, a header (and a trailer sometimes) is added at each protocol layer. During the de-encapsulation process, the headers (and/or trailers) are removed in reverse order. The user can step through the animation by clicking the "play" button repeatedly, or run the simulation continuously by checking the checkbox "Play continuously".

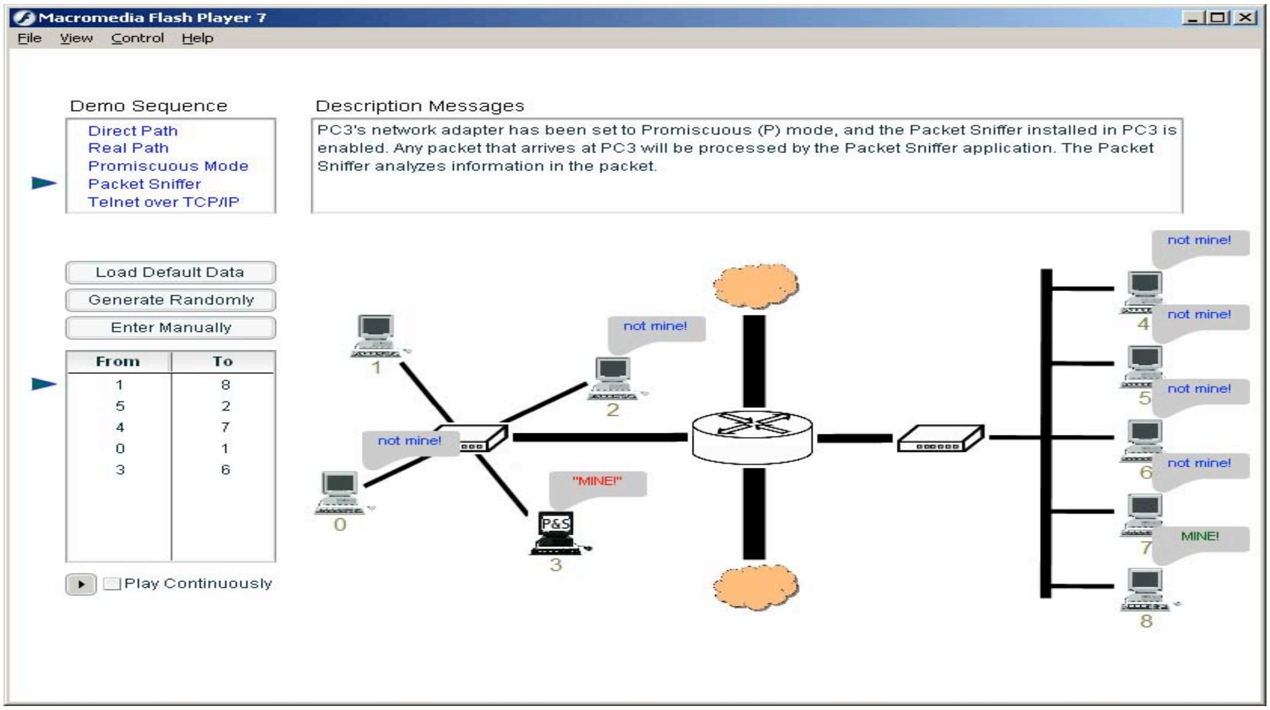

Figure 3. Demo IV: Packet sniffer

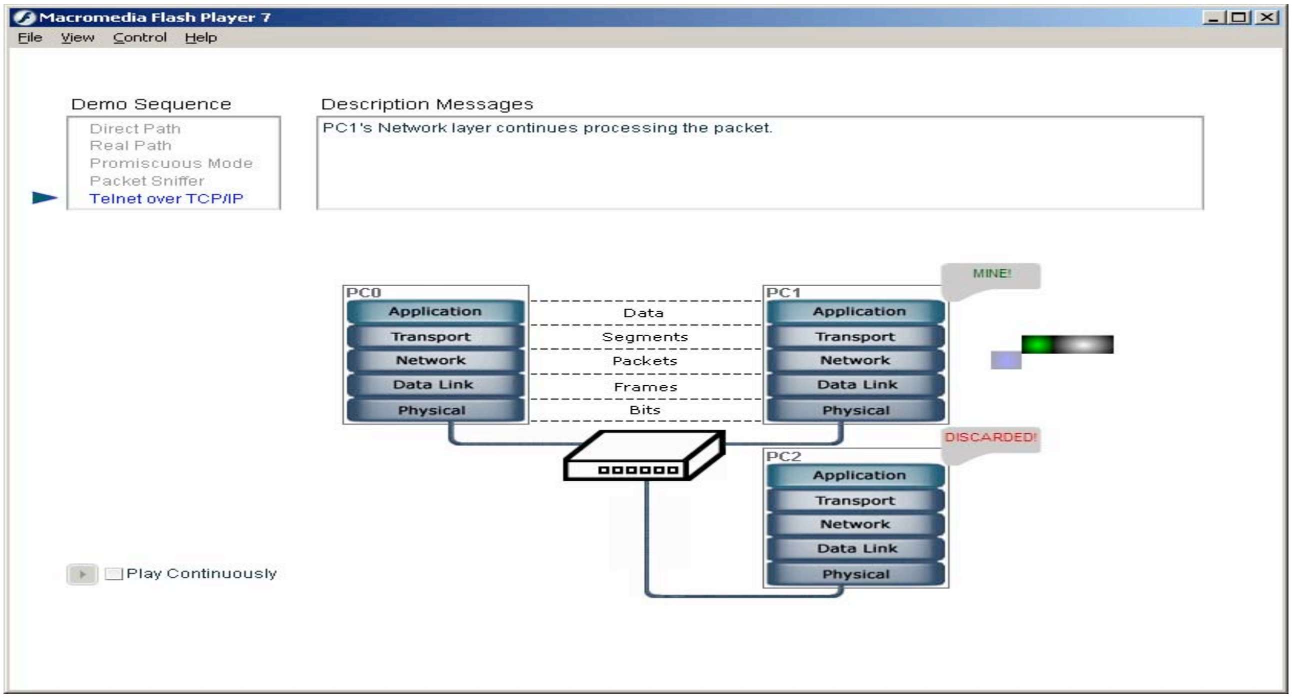

Figure 4. Demo V: The De-Encapsulation Process

In Figure 4, the data frame from PC0 is transmitted to PC1 and PC2. At the data link layer, the network interface hardware of PC2 checks the destination address of the incoming frame, and discards the data packet since its destination address it not PC2. PC1's network interface hardware recognizes that the data packet is addressed to PC1, so the frame is forwarded to the operating systems and is further de-encapsulated until it reaches the application layer in PC1.

4. EVALUATION

The packet sniffer simulator was used in the "Network Security" class in the Fall 2005 semester. There were 12 students in the class. Before the tool was introduced to the students, the students were first given a pre-test that reflects the following learning objectives:

a) Explain the differences between a hub, a bridge/switch, and a router
b) Explain bus and star topology
c) Explain how a data packet is transmitted in a local area network
d) Explain the purpose of "promiscuous mode" of a network interface
e) Explain what a packet sniffer does, and how it works.
f) Explain the encapsulation and de-encapsulation process of a data packet while going through the protocol stack

Then the instructor gave a brief introduction of the packet sniffer simulator in the class. The packet sniffer simulator is accessible through the web by the students. After the class, the students were given homework assignments on the packet sniffer simulator. Several weeks later, the students were given a post-test that reflects the above learning objectives of the packet sniffer simulator.

The pre-test and post-test scores of the students are shown in Figure 5. Figure 5 shows every student has done better in post-test than in pre-test. On average the students increased 36.7% from pre-test to post-test. This shows the packet sniffer simulator is effectiveness in helping the students in learning computer network and packet sniffer concepts.

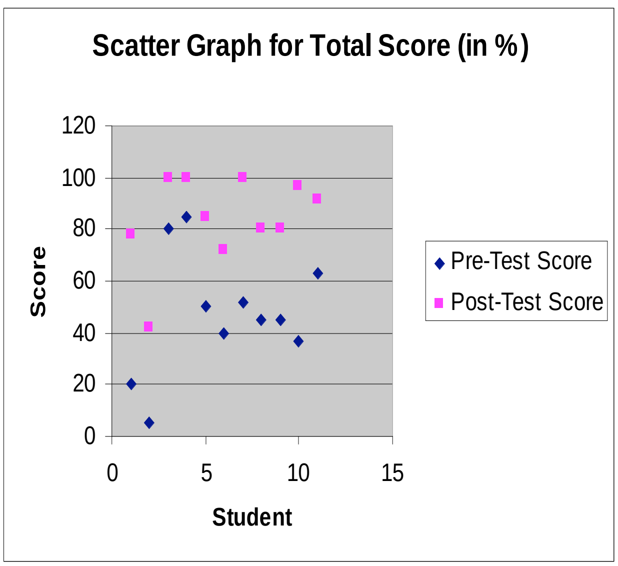

Figure 5. The Pre-Test and Post-Test scores of the students

A questionnaire was also given to the students after the tool is used. The result of the questionnaire is summarized in Table 1. Table 1 shows that 91.7% of the students agree that the packet sniffer simulator helped them in learning computer network and security concepts, and agree that the learning objectives are met by using the tool. 91.7% of the students would like to recommend this tool to others. All of the students in this class agree that the packet sniffer simulator is easy to learn and understand. We assume the students have various learning styles. However, all of the students would like to see more of this kind of visualization tools in the class they take.

The questionnaire result also shows that 83.34% of the students used the tool less than 4 times, and 83.34% of the students used the tool for less than 2 hours. We can conclude that the tool did not take a lot of time for most of the students, the overhead of learning this tool is low.

5. CONCLUSION

The packet sniffer animator demonstrates the packet sniffer related network concepts through a series of five demos. It can be used by instructors of computer networks and security in classroom. Tutorials and exercises are designed with the tool to help students understand the tool and engage them in the animation.

This tool was used in the "network security" course in Fall 2005 semester. A pre-test and post-test was given to the students before and after the tool was introduced to the students. Homework assignment on the tool was given to the students afterwards. Students opinions on the tool were also collected. The evaluation result shows that this animation tool is effective in meeting its learning objectives. Our future work includes refining this tool, developing new visualization tools for learning computer security concepts, and continuing the evaluation of this and other visualization tools for computer security.

6. ACKNOWLEDGEMENT

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Table 1. Student questionnaire result

	Strongly Agree	**Agree**	**Neither Agree or Disagree**	**Disagree**	**Strongly Disagree**
Do you think the demo helped in learning computer network and security concepts?	33.33%	58.33%	8.33%	0.0%	0.0%
Do you think the learning objectives are met by using the tool?	33.33%	58.33%	0.0%	8.33	0.0%
Do you think the tool helped you understand the questions asked in the homework?	25%	66.67%	8.33%	0.0%	0.0%
Do you think the web site and the tutorial were helpful in understanding the demo?	33.33%	66.67%	0.0%	0.0%	0.0%
Do you think this demo is easy to learn and understand?	50%	50%	0.0%	0.0%	0.0%
Would you like to see more of these demos (or simulators) in the remaining duration of this course?	66.67%	33.33%	0.0%	0.0%	0.0%
How likely are you to recommend this tool to others?	**Definitely will recommend**	**Probably will recommend**	**Not sure**	**Probably will not recommend**	**Definitely will not recommend**
	66.67%	25%	0.0%	8.33%	0.0%
How many times have used the demo?	**1 -- 2 times**	**2 -- 4 times**	**4 – 6 times**	**> 6 times**	
	16.67%	66.67%	0.0%	16.67%	
How much time did you spend on the demo?	**0 – 2 hours**	**2 – 4 hours**	**4 – 6 hours**	**6 – 8 hours**	**> 8 hours**
	83.33%	8.33%	0.0%	0.0%	8.33%